

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I _P. Ramesh (192312443) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Ramesh

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- Ii. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- Iii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) $P3: \neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1)ANS;

Smart Medical Diagnosis System

a)Representation using Propositional Logic

Step 1: Define the symbols

For Patient A, let:

- F = Patient A has Fever
- C = Patient A has Cough
- R = Patient A has Rash
- B = Patient A has Breathlessness
- Flu = Patient A has Possible Flu
- M = Patient A has Possible Measles
- P = Patient A has Possible Pneumonia
-

Step 2: Represent the rules

Rule I: Fever AND Cough $\rightarrow$ Possible Flu

Rule II: Fever AND Rash $\rightarrow$ Possible Measles

Rule III: Cough AND Breathlessness $\rightarrow$ Possible Pneumonia

Step 3: Represent the facts

Given facts for Patient A:

There is no fact that Patient A has Rash, so:

Symbols Used

Symbol	Meaning
$\wedge$	AND
$\rightarrow$	Implies / If-Then
True	Fact is known to be true
F	Fever
C	Cough
R	Rash
B	Breathlessness
Flu	Possible Flu
M	Possible Measles

Symbol	Meaning
P	Possible Pneumonia

(b) Forward Chaining

Forward chaining starts with the known facts and applies rules whose conditions are satisfied.

Initial Facts

That is:

- Fever = True
- Cough = True
- Breathlessness = True

Inference Step 1 — Rule I

Rule I:

We have:

Therefore, Rule I fires.

Derived Fact:

Diagnosis: Possible Flu.

Inference Step 2 — Rule II

Rule II:

We have:

- Fever = True
- Rash = not given

Inference Step 3 — Rule III

Rule III:

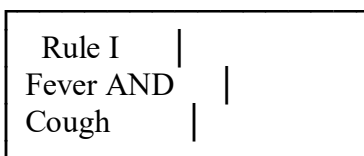
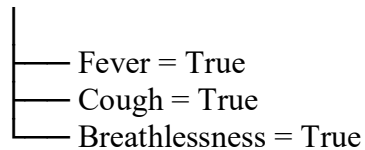
We have:

Therefore, Rule III fires.

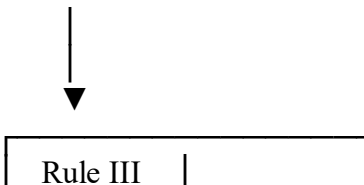
.

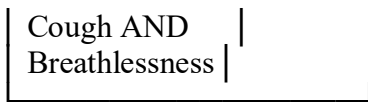
Forward Chaining Summary

Initial Facts



Possible Flu





Possible Pneumonia

Final Diagnoses

Diagnosis	Result	Reason
Possible Flu	Derived	Fever + Cough
Possible Measles	Not derived	Rash unavailable
Possible Pneumonia Derived Cough + Breathlessness		

Therefore:

(c) Backward Chaining for Pneumonia

The goal is:

Backward chaining starts with the goal and works backward to find rules that can prove it.

Step 1: Start with the Goal

Goal: Patient A has Pneumonia

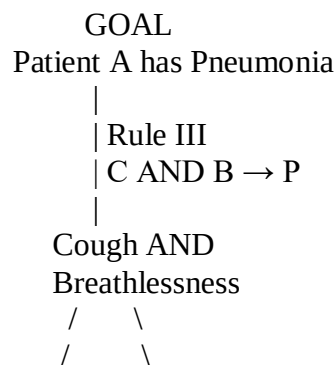
Find a rule whose conclusion is Pneumonia.

Rule III:

Therefore, to prove Pneumonia, we need to prove:

- Cough
- Breathlessness

Goal Reduction Tree



Cough=True Breathlessness=True

FACT FACT

2)ANS:

a)Unification of Rule 1

Rule 1 is:

To apply the rule, we match its premises with the facts.

Step 1: Match Vehicle(x)

Rule premise:

Fact:

Therefore:

Step 2: Match EmergencyType(x)

After applying the substitution:

Given fact:

It matches exactly.

Therefore:

Most General Unifier (MGU)

Applying this substitution to Rule 1:

Since both premises are true:

is derived.

(b) Resolution Proof of Proceed(CarA)

We want to prove:

Using resolution, we first convert the relevant rules into **CNF**.

Step 1: Convert Rule 1 to CNF

Original:

Using:

Therefore:

Step 2: Convert Rule 2 to CNF

Original:

This gives two clauses:

and

Step 3: Convert Rule 3 to CNF

Original:

Therefore:

Step 4: Ground the Rules

From the facts:

and

Rule 1 gives:

Using Rule 2:

Also:

Step 5: Resolution Derivation

Resolution 1

Clause:

Facts:

Resolving them gives:

Resolution 2

Rule 2 clause:

and:

Resolution gives:

Resolution 3

Rule 3:

Given facts:

Resolve successively:

Then:

Then:

Therefore:

Resolution Flow

Vehicle(Ambulance)

+

EmergencyType(Ambulance)



ClearPath(Ambulance)



GreenSignal(Ambulance)



Vehicle(CarA)

Behind(CarA,Ambulance)

GreenSignal(Ambulance)



Proceed(CarA)

(c) Handling Two Simultaneous Emergency Vehicles Is the current KB sufficient?

Partially yes, but not completely.

The existing rules use universal variables, so they can technically apply to multiple emergency vehicles independently.

For example, if another emergency vehicle FireTruck is added:

Rule 1 gives:

Rule 2 gives:

So the basic emergency-vehicle processing works.

However, there is a limitation

The current KB does not specify how to handle conflicts between two simultaneous emergency vehicles.

For example:

Ambulance → approaching from Road 1

FireTruck → approaching from Road 2

Both may require a green signal at the same intersection. The current KB does not specify:

- Which emergency vehicle has priority?
- Which road should receive the green signal?
- How conflicting green signals are prevented.
- How vehicles are queued.
- How emergency priority is determined.

3)ANS:

Agricultural Expert Reasoning System

Given Knowledge Base

1. **P1:** $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
2. **P2:** $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
3. **P3:** $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
4. **Facts:** $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

Let:

- **S** = SoilDry
- **I** = IrrigationNeeded
- **W** = CropWheat
- **D** = ApplyDripMethod
- **R** = CropAtRisk

(a) Conversion into CNF

P1: SoilDry $\rightarrow$ IrrigationNeeded

Original

Eliminate implication

Rule:

Therefore:

CNF

P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Original

Eliminate implication

Apply De Morgan's Law

Therefore:

This is already in CNF.

P3: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Original

Eliminate implication

Move negation inward

Using De Morgan's Law:

Since:

we get:

This is already in CNF.

Facts in CNF

Fact 1

Therefore:

Fact 2

Therefore:

Complete CNF Knowledge Base

Clause	CNF
P1	

Clause	CNF
P2	
P3	
Fact 1	
Fact 2	

(b) Resolution Refutation to Prove ApplyDripMethod Goal

We want to prove:

In Resolution Refutation, negate the goal.

Negated Goal

Add it to the CNF knowledge base.

Step 1: Derive IrrigationNeeded

P1:

Fact:

Resolve them:

$\neg S \vee I$

S

I

Therefore:

Step 2: Use P2

P2:

We derived:

Resolution:

$\neg I \vee \neg W \vee D$

I

$\neg W \vee D$

Therefore:

Step 3: Use CropWheat Fact

We have:

Resolve with:

$\neg W \vee D$

W

D

Therefore:

Step 4: Resolve with Negated Goal

Negated goal:

We have derived:

Resolution:

D

$\neg D$

□

The result is the **empty clause**:

Therefore, the negated goal leads to a contradiction.

Hence:

Important observation

P3 is not needed to prove ApplyDripMethod. It only becomes relevant when determining whether CropAtRisk can be derived.

(c) Remove the Fact SoilDry

Now remove:

The remaining facts are:

The rules remain:

Step 1: Can IrrigationNeeded be derived

P1 says:

But now:

Therefore, we **cannot derive I**.

Step 2: Can ApplyDripMethod be derived

P2 requires:

We have:

but:

Therefore:

Step 3: Can CropAtRisk be derived

P3 is:

To derive CropAtRisk, we need:

and:

We know:

But we do not know:

The absence of D does not mean $\neg D$.

Therefore:

4)ANS:

Wumpus World Knowledge Agent

Given Percepts

The agent perceives:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Rules

1. Stench [1,2] $\rightarrow$ Wumpus is adjacent to [1,2]
2. Breeze [1,1] $\rightarrow$ Pit is adjacent to [1,1]
3. Glitter [x,y] $\rightarrow$ Gold is at [x,y]
4. $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Belief State and Modus Ponens

Step 1: Initial Percept Facts

The agent's initial belief state is:

Step 2: Apply Rule 1

Rule:

We know:

By Modus Ponens:

The cells adjacent to [1,2] are:

Therefore, the Wumpus could be in one of these cells.

Belief State Summary

Percept/Conclusion	Status
Stench[1,2]	Known
Breeze[1,1]	Known
Glitter[2,2]	Known
Wumpus adjacent to [1,2]	Derived
Pit adjacent to [1,1]	Derived
Gold at [2,2]	Derived
Safe cells	Not established

(b) Forward Chaining: Possible Wumpus and Pit Locations

Forward chaining starts from the known percepts and repeatedly applies applicable rules.

Wumpus Reasoning

Initial fact

Apply Rule 1:

Therefore:

The adjacent cells are:

[1,3]
|
[1,1] — [1,2] — [1,3]
|
[2,2]

Pit Reasoning

Initial fact:

Apply Rule 2:

The cells adjacent to [1,1] are:

Therefore, possible Pit locations are:

Gold Reasoning

Initial fact:

Apply Rule 3:

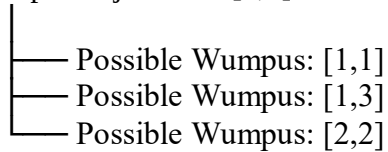
Therefore:

Forward-Chaining Summary

Stench[1,2]



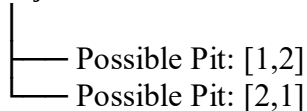
Wumpus adjacent to [1,2]



Breeze[1,1]



Pit adjacent to [1,1]



Glitter[2,2]



Gold[2,2]

(c) Is the Path [1,1] → [1,2] → [2,2] Safe

The proposed path is:

The agent wants to reach the gold at [2,2].

Cell [1,1]

There is no direct evidence that [1,1] contains a pit, but it is a possible Wumpus location because [1,1] is adjacent to the stench at [1,2].

Therefore, [1,1] is not proven safe.

Cell [1,2]

This cell has:

Also, because of the Breeze at [1,1]:

is a possible Pit location.

Therefore:

The agent cannot establish that [1,2] is safe.

Cell [2,2]

The agent knows:

However, [2,2] is also adjacent to [1,2], where a stench is perceived.

Therefore:

So the destination containing the gold is not proven safe.